CSBS CAPSTONE · ES · ITPM · ASTMA

Sentilink Enterprise: Book Review Sentiment Analysis & Decision Support Platform

Unifying Natural Language Processing, 3-Tier Enterprise Systems, and IT Project Management

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Academic Tri-Subject Foundation

How three engineering disciplines converge to transform reader feedback into publishing intelligence.

1 ASTMA (Elective IV B)

Text Mining & NLP Engine. Preprocessing literary feedback, polarity scoring, and RAKE keyword extraction.

2 Enterprise Systems

3-Tier Architecture & BI. Scalable FastAPI services, relational SQLite database, RBAC security, and BI analytics.

3 IT Project Management

Execution & Governance. 4-sprint Agile scheduling, PERT critical path analysis, and budget control.

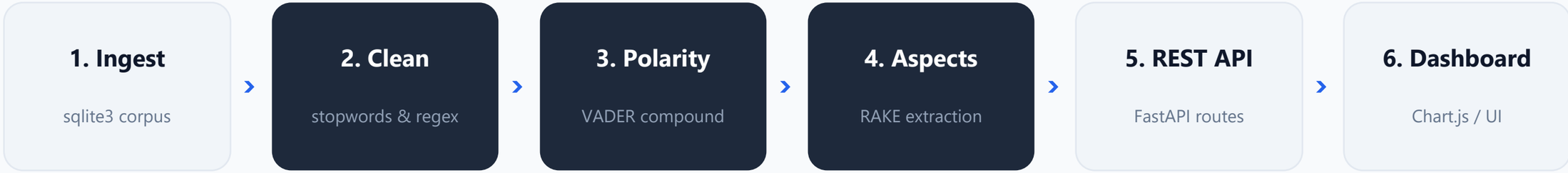
36 Master Books in Catalog

35 Literary Genres Filtered

124+ Verified Reviews Analysed

The Sentilink Text Analytics Pipeline

A modular, stage-by-stage architecture turning raw reader prose into actionable business intelligence.



Highlighted dark stages represent the core ASTMA processing pipeline.

Why Sequential Decoupling?

Each stage isolates concerns: rigorous cleaning eliminates vocabulary skew without losing valence cues, sentiment scores feed aspect indexing, and asynchronous REST endpoints allow the dashboard to render instantly.

Text Preprocessing & Normalization

Refining literary reviews to isolate semantic features while preserving emotional intensity cues.

1. **Case Normalization**
   Unifies 'Masterpiece' and 'masterpiece' into identical tokens

2. **Noise & Symbol Removal**
   Filters web URLs, HTML entities, and formatting symbols

3. **Word Tokenization**
   Segments continuous literary prose into discrete lexical units

4. **Custom Stopword Pruning**
   Removes generic terms ('the', 'page') while preserving valence modifiers

5. **Morphological Lemmatization**
   Maps inflected variations ('reading', 'reads') to base lemma 'read'

RAW READER REVIEW (BEFORE)

The protagonist's character arc was truly magnificent, but the slow pacing in chapter 8 made me lose interest!

EXTRACTED FEATURE TOKENS (AFTER)

protagonist character arc truly magnificent slow pacing chapter 8 lose interest

Sentiment Classification Paradigms

Evaluating NLP methodologies for literary review analytics: why a hybrid approach was chosen.

Lexicon-Based (VADER)

Evaluates sentiment using curated word valence dictionaries.

Strength: Zero training data, instant explainability

Limit: Insensitive to deep literary sarcasm

Primary Production Engine

Statistical ML (TF-IDF)

Trains Naive Bayes & Logistic Regression on n-gram vectors.

Strength: Learns book-specific literary vocabularies

Limit: Requires verified labels

Evaluated as Baseline

Deep Transformers

Bidirectional contextual embeddings (BERT / RoBERTa).

Strength: Understands complex metaphors & tone

Limit: High RAM & latency

Future Scale Target

Polarity Scoring & Valence Heuristics

How grammatical cues and modifier heuristics calculate the normalized compound sentiment.

1. **Punctuation Stacking** 'Magnificent!' vs 'Magnificent!!!' raises intensity
2. **Capitalization Emphasis** 'MASTERPIECE' boosts positive polarity score
3. **Degree Modifiers** 'Very engaging' raises score; 'barely readable' lowers
4. **Polarity Inversion** 'Not captivating' inverts positive word to negative va
5. **Contrastive Conjunctions** 'Good start, BUT dull ending' weights concl

Compound Metric Scale: -1.0 to +1.0



The engine computes raw positive, neutral, and negative word proportions, and categorizes the review using the normalized compound score calculated from valence adjustments.

Aspect-Based Opinion Mining for Books

Dissecting reader reviews into 8 literary craft dimensions to discover WHY readers loved or disliked a book.

- **Plot & Narrative**
- **Character Arc**
- **Pacing & Rhythm**
- **Writing Style**
- **World Building**
- **Emotional Impact**
- **Theme & Tone**
- **Ending & Resolution**

READER REVIEW

The world building was breathtaking, but the characters felt shallow and the ending was rushed.

MINED OPINION DIMENSIONS

- World Building: Positive (+0.82) — 'breathtaking'
- Character Arc: Negative (-0.45) — 'shallow'
- Ending & Resolution: Negative (-0.61) — 'rushed'

3-Tier Enterprise Architecture

Decoupled Service-Oriented Architecture (SOA) ensuring performance, maintainability, and security.

Presentation Tier (UI)

Responsive Web Dashboard, Chart.js Visualizations, Genre Filters

Application & Logic Tier (SOA)

FastAPI Microservices, VADER Sentiment Engine, RAKE Extractor

Data Persistence Tier

SQLite relational database: books, reviews, users, audit logs

Enterprise Characteristics

- Unified Data Platform: Consolidates catalog data and NLP analytics into a single dashboard.
- Role-Based Access Control: Granular security separating Admin controls from Reader views.
- SOA Interoperability: Modular REST endpoints ready for external e-commerce integration.

Relational Database & Decision Support

Structured data engineering powering publishing market analysis and reader satisfaction metrics.

1 Relational Schema Design

Normalized SQLite schema linking Books, Reviews, Users, and Genres with referential integrity.

2 B-Tree Query Indexing

Indexes on book_id, sentiment_class, and genre ensure sub-10ms query execution across reviews.

3 Star vs Sentiment Divergence

Detects reviews where reader gave 4/5 stars but NLP text score reveals severe narrative criticisms.

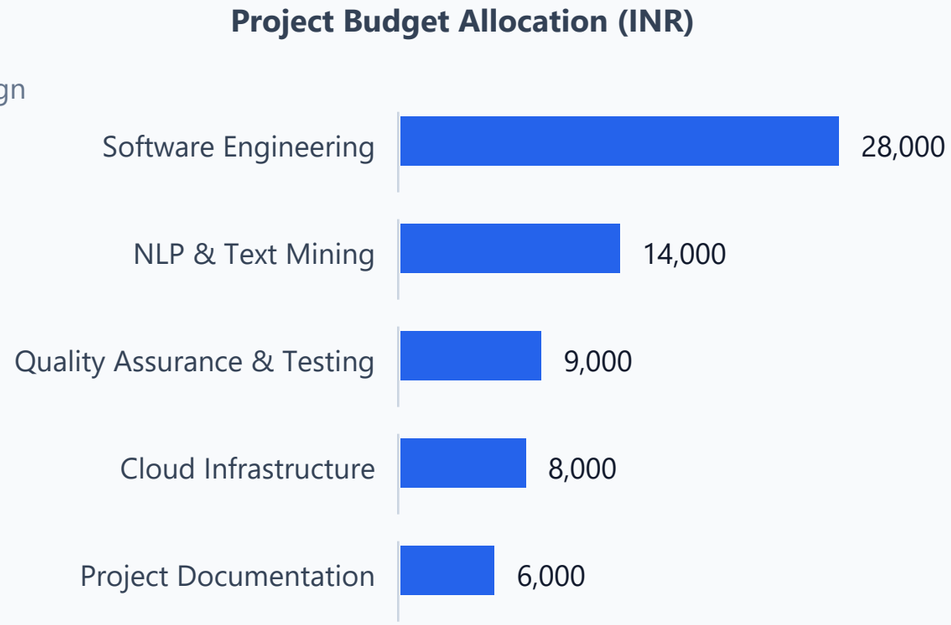
4 Genre Satisfaction Index

Computes aggregate sentiment across 35 literary genres to identify high-performing literary categories.

IT Project Management & Agile Delivery

Executing the project using an 8-week Scrum methodology with budget and schedule control.

1. **Sprint 1 (W1-2)** Corpus Ingestion (36 books) & Database Schema Design
2. **Sprint 2 (W3-4)** NLP Pipeline, VADER Engine & RAKE Aspect Discovery
3. **Sprint 3 (W5-6)** FastAPI Microservices, Analytics Dashboard & RBAC
4. **Sprint 4 (W7-8)** Integration Testing, Crashing Analysis & Cloud Launch



Rs. 65,000 Total Project Expenditure

Risk Management & Quality Assurance

Proactive identification of technical and operational risks with targeted mitigation strategies.

Risk Description	Probability	Impact	Mitigation Strategy
Literary Idiom Misinterpretation	Medium	Medium	Augment lexicon with domain-specific literary modifiers
Database Write Concurrency	Low	High	Implement SQLite WAL mode and serialized connection pool
Sprint Scope Creep	Medium	High	Enforce strict Scrum backlog freezes and milestone reviews
Corpus Ingestion Format Errors	Low	Medium	Implement strict Pydantic input models and validation hooks
Cloud Deployment Cold-Start	Medium	Low	Configure health ping heartbeat to keep background services warm
Subjectivity Ambiguity in Reviews	Medium	Medium	Combine polarity scores with RAKE aspect phrase co-occurrence

Technical Evaluation & Known Limitations

Academic honesty: examining current design trade-offs and algorithmic constraints.

1 Document-Level Aggregation

Each review produces one compound score, which blends mixed aspect opinions into an average.

2 Irony & Sarcasm Boundaries

Subtle literary sarcasm (e.g. 'What a joy to read 500 pages of filler') requires attention transformers.

3 Classical Prose Vocabulary

Standard modern lexicons occasionally misread archaic 19th-century adjectives in classical fiction.

4 Proxy Star-Rating Validation

Validation relies on star ratings as proxy ground truth rather than crowd-annotated sentiment tags.

Project Conclusion & Future Roadmap

Delivered Capabilities

- End-to-end 3-Tier Enterprise system deployed live with FastAPI and SQLite
- Dual sentiment engine combining VADER polarity and RAKE aspect extraction
- Interactive BI dashboard cataloging 36 books across 35 literary genres
- Managed under Agile Scrum with CPM activity crashing and budget tracking

Future Engineering Roadmap

- Fine-tune domain-specific BERT / DeBERTa models for sentence-level ABSA
- Mobile barcode scanner integration for in-store instant book sentiment lookups
- Reader personalization engine combining sentiment profiles with collaborative filtering
- Cross-lingual sentiment mining for regional Indian and translated literature